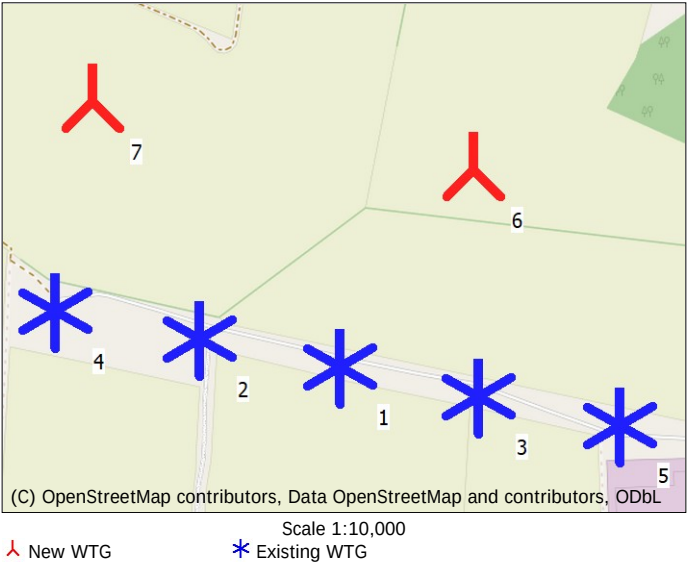


DECIBEL - Main Result

Calculation: 5 Old WTG and 2 New WTG

Noise propagation model:
ISO 9613-2:2024 United Kingdom
Wind speed:
4.0 m/s - 12.0 m/s, step 1.0 m/s
Ground attenuation:
General, Ground factor: 0.5
Valley effect, Penalty: 3.0 dB
Topographic screening, Reduction: 2.0 dB
Type of demand in calculation:
WTG noise is compared to ambient noise plus 5dB margin with the option of a floor setting (e.g. 35dB)
Noise values in calculation:
All noise values are 90% exceedence values (L90) designed to show compliance with ETSU-R-97 limits
Pure tones:
Fixed penalty added to source noise of WTGs with pure tones
WTG catalogue
Uncertainty margin:
0.0 dB; Uncertainty margin in NSA has priority
Calculation height above ground level:
4.0 m
Octave band data required

All coordinates are in
British TM-OSGB36/Airy (GB/IE)



WTGs

	Easting	Northing	Z	Row data/Description	WTG type			Power, rated	Rotor diameter	Hub height	Noise data		First wind speed [m/s]	LwaRef [dB(A)]	Last wind speed [m/s]	LwaRef [dB(A)]
					Valid	Manufact.	Type-generator				Creator	Name				
			[m]					[kW]	[m]	[m]						
1	424,115	191,390	109.3	BONUS 1.3 MW 1300-25...	No	BONUS	1.3 MW-1,300/250	1,300	62.0	50.0	EMD	Level 0 - - octave - 06-2003	4.0	98.0	12.0	106.0 g
2	423,930	191,427	112.1	BONUS 1.3 MW 1300-25...	No	BONUS	1.3 MW-1,300/250	1,300	62.0	50.0	EMD	Level 0 - - octave - 06-2003	4.0	98.0	12.0	106.0 g
3	424,300	191,353	107.9	BONUS 1.3 MW 1300-25...	No	BONUS	1.3 MW-1,300/250	1,300	62.0	50.0	EMD	Level 0 - - octave - 06-2003	4.0	98.0	12.0	106.0 g
4	423,738	191,463	113.3	BONUS 1.3 MW 1300-25...	No	BONUS	1.3 MW-1,300/250	1,300	62.0	50.0	EMD	Level 0 - - octave - 06-2003	4.0	98.0	12.0	106.0 g
5	424,488	191,315	106.0	BONUS 1.3 MW 1300-25...	No	BONUS	1.3 MW-1,300/250	1,300	62.0	50.0	EMD	Level 0 - - octave - 06-2003	4.0	98.0	12.0	106.0 g
6	424,293	191,652	103.2	VESTAS V162-6.2 6200 1...	Yes	VESTAS	V162-6.2-6,200	6,200	162.0	119.0	EMD	PO6200 - 2024-10	4.0	97.3	12.0	104.8 g
7	423,787	191,740	113.4	ENERCON E-160 EP5 46...	No	ENERCON	E-160 EP5-4,600	4,600	160.0	119.9	EMD	Mode 00 - OM 0 s (4600 kW)	4.0	100.1	12.0	106.2
g) Data calculated from data for other wind speed (uncertain)																

DECIBEL - Assumptions for noise calculation

Calculation: 5 Old WTG and 2 New WTG

Noise calculation model:
ISO 9613-2:2024 United Kingdom
Wind speed (at 10 m height):
4.0 m/s - 12.0 m/s, step 1.0 m/s
Ground attenuation:
General, Ground factor: 0.5
Valley effect, Penalty: 3.0 dB
Topographic screening, Reduction: 2.0 dB
Meteorological coefficient, C0:
Selected option: Fixed value: 0.0 dB
Type of demand in calculation:
3: WTG noise is compared to ambient noise plus margin (UK, AT etc.)
Noise values in calculation:
All noise values are 90% exceedance values (L90)
Pure tones:
Fixed penalty added to source noise of WTGs with pure tones
WTG catalogue
Height above ground level, when no value in NSA object:
4.0 m; Don't allow override of model height with height from NSA object
Uncertainty margin:
0.0 dB; Uncertainty margin in NSA has priority
Deviation from "official" noise demands. Negative is more restrictive, positive is less restrictive.:
0.0 dB(A)
Octave data required
Frequency dependent air absorption
63 125 250 500 1,000 2,000 4,000 8,000
[dB/km] [dB/km] [dB/km] [dB/km] [dB/km] [dB/km] [dB/km] [dB/km]
0.1 0.4 1.0 1.9 3.7 9.7 32.8 117.0

The air absorption is for a temperature of 10.0 degrees C and 70.0 % humidity.

All coordinates are in
British TM-OSGB36/Airy (GB/IE)

WTG: VESTAS V162-6.2 6200 162.0 !O!
Noise: PO6200 - 2024-10

Source Source/Date Creator Edited
Vestas 2024-10-31 EMD 2024-11-27 3:17 PM
DMS no.: 0180-3964 V00.

Status	Hub height	Wind speed	LwA.ref	Pure tones	Octave data							
					63	125	250	500	1000	2000	4000	8000
	[m]	[m/s]	[dB(A)]		[dB]	[dB]	[dB]	[dB]	[dB]	[dB]	[dB]	[dB]
From Windcat	119.0	4.0	97.3	No	88.8	90.1	89.6	89.5	89.3	88.9	82.1	69.6
From Windcat	119.0	5.0	101.0	No	85.2	92.3	94.6	93.5	94.2	93.5	86.5	74.3
From Windcat	119.0	6.0	103.9	No	87.6	94.6	96.6	96.4	97.6	97.2	91.0	80.1
From Windcat	119.0	7.0	104.8	No	88.4	96.3	97.2	97.2	98.4	97.8	91.8	82.2
From Windcat	119.0	8.0	104.8	No	87.4	95.2	96.8	97.6	98.7	98.0	92.4	84.1
From Windcat	119.0	9.0	104.8	No	86.9	95.1	96.7	97.7	98.8	98.0	92.2	84.2
From Windcat	119.0	10.0	104.8	No	87.8	95.5	97.1	97.5	98.7	97.8	91.8	83.1
Extrapolated	119.0	11.0	104.8	No	86.9	95.1	96.7	97.7	98.8	98.0	92.3	84.2
Extrapolated	119.0	12.0	104.8	No	87.8	95.5	97.1	97.6	98.7	97.9	91.8	83.1



Project: GB_Project Description: Demo from youtube (user: ReNewAbility Academy) using United Kingdom farm site

Licensed user: Adam Nearing
The license can only be used for educational purposes
--

Adam Nearing / w0485009@nsc.ca
Calculated:
2026-03-14 3:42 PM/4.2.285

DECIBEL - Assumptions for noise calculation

Calculation: 5 Old WTG and 2 New WTG

WTG: ENERCON E-160 EP5 4600 160.0 !O!

Noise: Mode 00 - OM 0 s (4600 kW)

Source Source/Date Creator Edited
ENERCON GmbH 2022-09-09 EMD 2022-09-09 2:00 PM

The sound power levels do not include uncertainties.

According to manufacturer specification document

(D0799705-6_#_de_#_Datenblatt_Betriebsmodus_E-160_EP5_4600_kW_mit_TES.pdf/D0820550-3_#_de_#_Datenblatt_Terzbandpegel_E-160_EP5_4600_kW_mit_TES.pdf).

Enercon reserves the right to change the above specifications without prior notice.

Status	Hub height [m]	Wind speed [m/s]	LwA,ref [dB(A)]	Pure tones	Octave data							
					63	125	250	500	1000	2000	4000	8000
From Windcat	119.9	4.0	100.1	No	80.8	86.2	88.8	92.3	94.6	95.4	89.3	72.1
From Windcat	119.9	5.0	104.6	No	84.8	90.3	92.8	96.3	99.2	100.2	94.2	77.0
From Windcat	119.9	6.0	105.8	No	85.8	91.5	94.2	97.8	100.4	101.3	95.3	78.1
From Windcat	119.9	7.0	106.0	No	86.1	91.6	93.8	97.1	100.5	101.9	96.0	78.7
From Windcat	119.9	8.0	106.2	No	86.6	92.0	94.1	96.9	100.1	102.3	96.7	79.7
From Windcat	119.9	9.0	106.2	No	86.7	92.2	94.4	97.3	100.0	102.1	96.8	80.0
From Windcat	119.9	10.0	106.2	No	86.8	92.3	94.7	97.6	100.1	102.0	96.8	80.2
From Windcat	119.9	11.0	106.2	No	86.8	92.4	94.9	97.8	100.1	101.8	96.7	80.2
From Windcat	119.9	12.0	106.2	No	86.9	92.6	95.1	98.0	100.1	101.7	96.6	80.2

WTG: BONUS 1.3 MW 1300-250 62.0 !O!

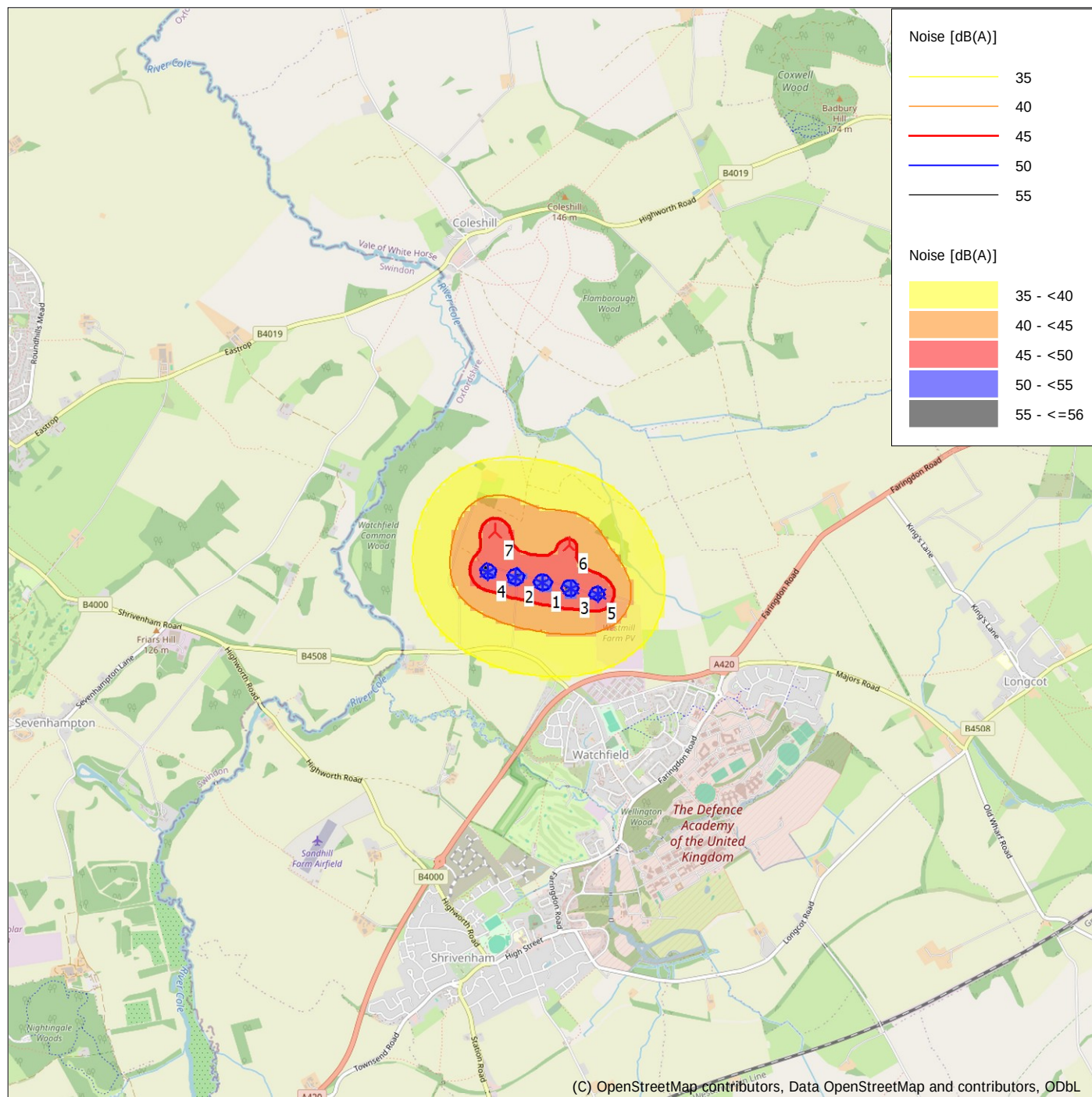
Noise: Level 0 - - octave - 06-2003

Source Source/Date Creator Edited
Manufacturer 2003-06-30 EMD 2005-06-24 2:53 PM

Status	Hub height [m]	Wind speed [m/s]	LwA,ref [dB(A)]	Pure tones	Octave data							
					63	125	250	500	1000	2000	4000	8000
From slope	50.0	4.0	98.0	No	From nearest other wind speed				82.1	88.7	91.6	90.7
Interpolated	50.0	5.0	99.0	No	Generic data				80.6	87.6	91.0	93.6
Interpolated	50.0	6.0	100.0	No	Generic data				81.6	88.6	92.0	94.6
Interpolated	50.0	7.0	101.0	No	Generic data				82.6	89.6	93.0	95.6
From Windcat		8.0	102.0	No					86.1	92.7	95.6	94.7
Extrapolated	50.0	9.0	103.0	No	Generic data				84.6	91.6	95.0	97.6
Extrapolated	50.0	10.0	104.0	No	Generic data				85.6	92.6	96.0	98.6
Extrapolated	50.0	11.0	105.0	No	Generic data				86.6	93.6	97.0	99.6
Extrapolated	50.0	12.0	106.0	No	Generic data				87.6	94.6	98.0	100.6

DECIBEL - Map 4.0 m/s

Calculation: 5 Old WTG and 2 New WTG



0 500 1000 1500 2000 m

Map: EMD OpenStreetMap, Print scale 1:40,000, Map center British TM-OSGB36/Airy (GB/IE) East: 424,113 North: 191,527

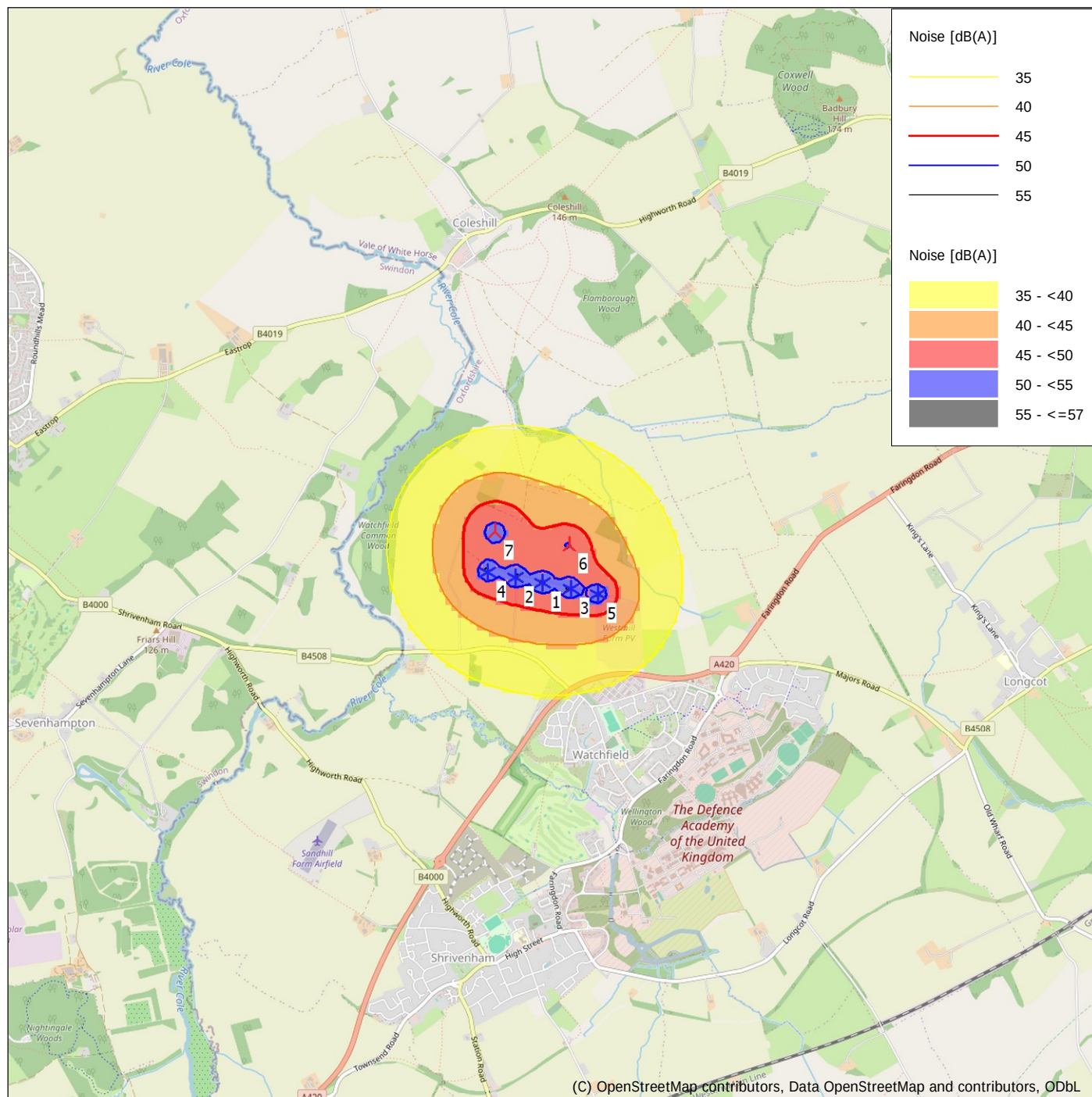
New WTG

Existing WTG

Noise calculation model: ISO 9613-2:2024 United Kingdom. Wind speed: 4.0 m/s
Height above sea level from active line object

DECIBEL - Map 5.0 m/s

Calculation: 5 Old WTG and 2 New WTG



New WTG

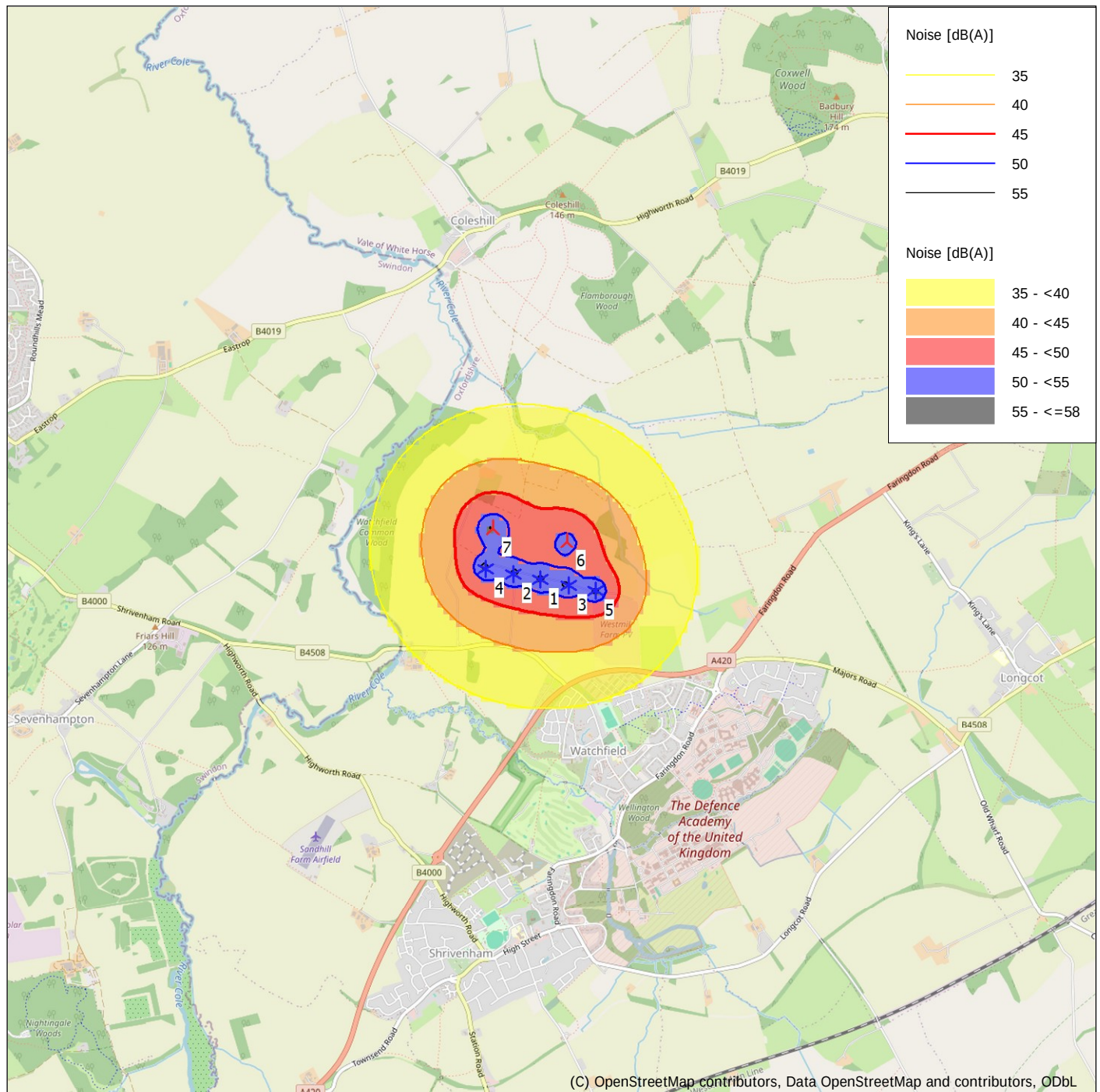
Map: EMD OpenStreetMap, Print scale 1:40,000, Map center British TM-OSGB36/Airy (GB/IE) East: 424,113 North: 191,527

Existing WTG

Noise calculation model: ISO 9613-2:2024 United Kingdom. Wind speed: 5.0 m/s
Height above sea level from active line object

DECIBEL - Map 6.0 m/s

Calculation: 5 Old WTG and 2 New WTG



0 500 1000 1500 2000 m

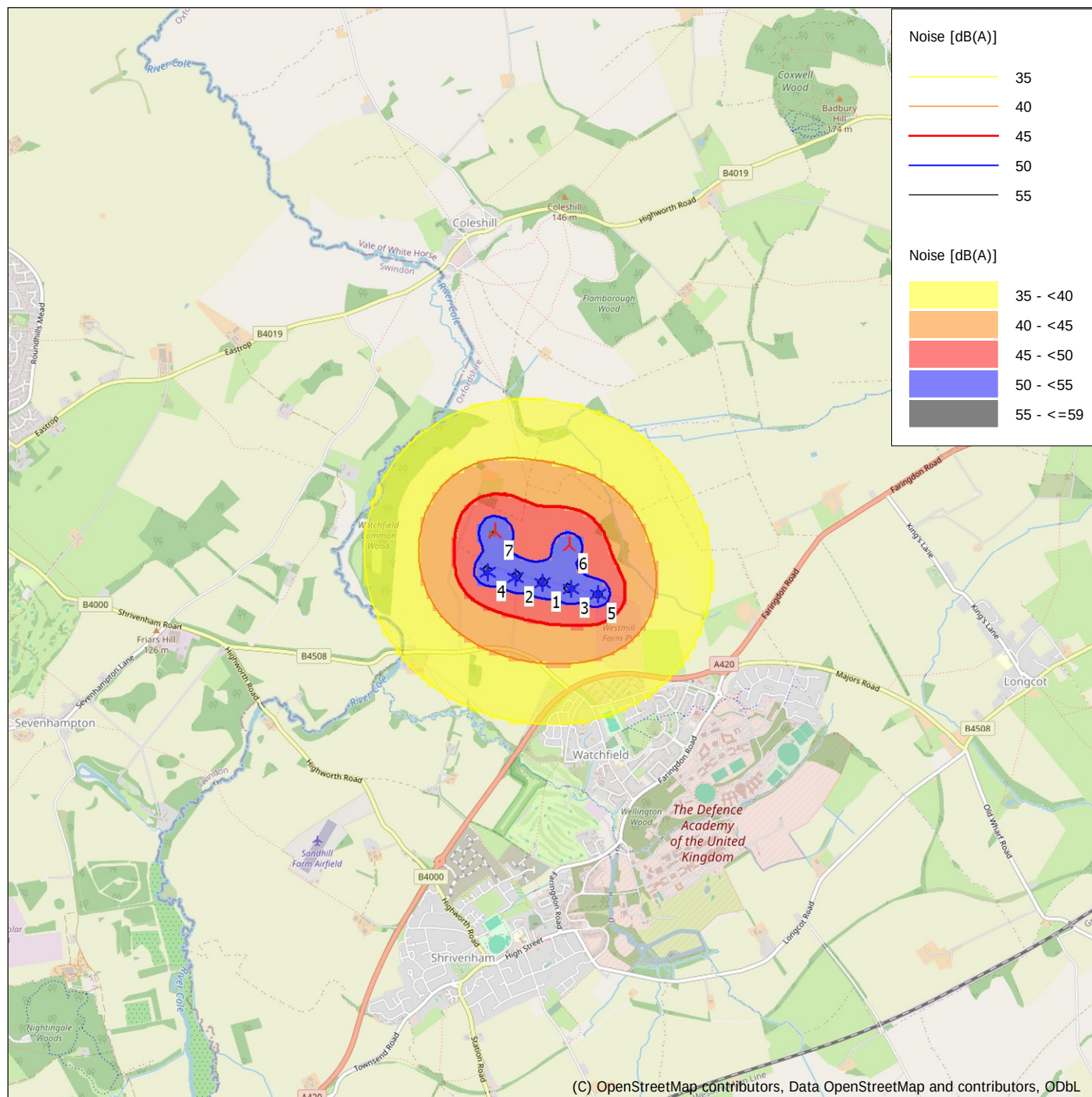
Map: EMD OpenStreetMap, Print scale 1:40,000, Map center British TM-OSGB36/Airy (GB/IE) East: 424,113 North: 191,527

New WTG Existing WTG

Noise calculation model: ISO 9613-2:2024 United Kingdom. Wind speed: 6.0 m/s
Height above sea level from active line object

DECIBEL - Map 7.0 m/s

Calculation: 5 Old WTG and 2 New WTG



0 500 1000 1500 2000 m

Map: EMD OpenStreetMap, Print scale 1:40,000, Map center British TM-OSGB36/Airy (GB/IE) East: 424,113 North: 191,527

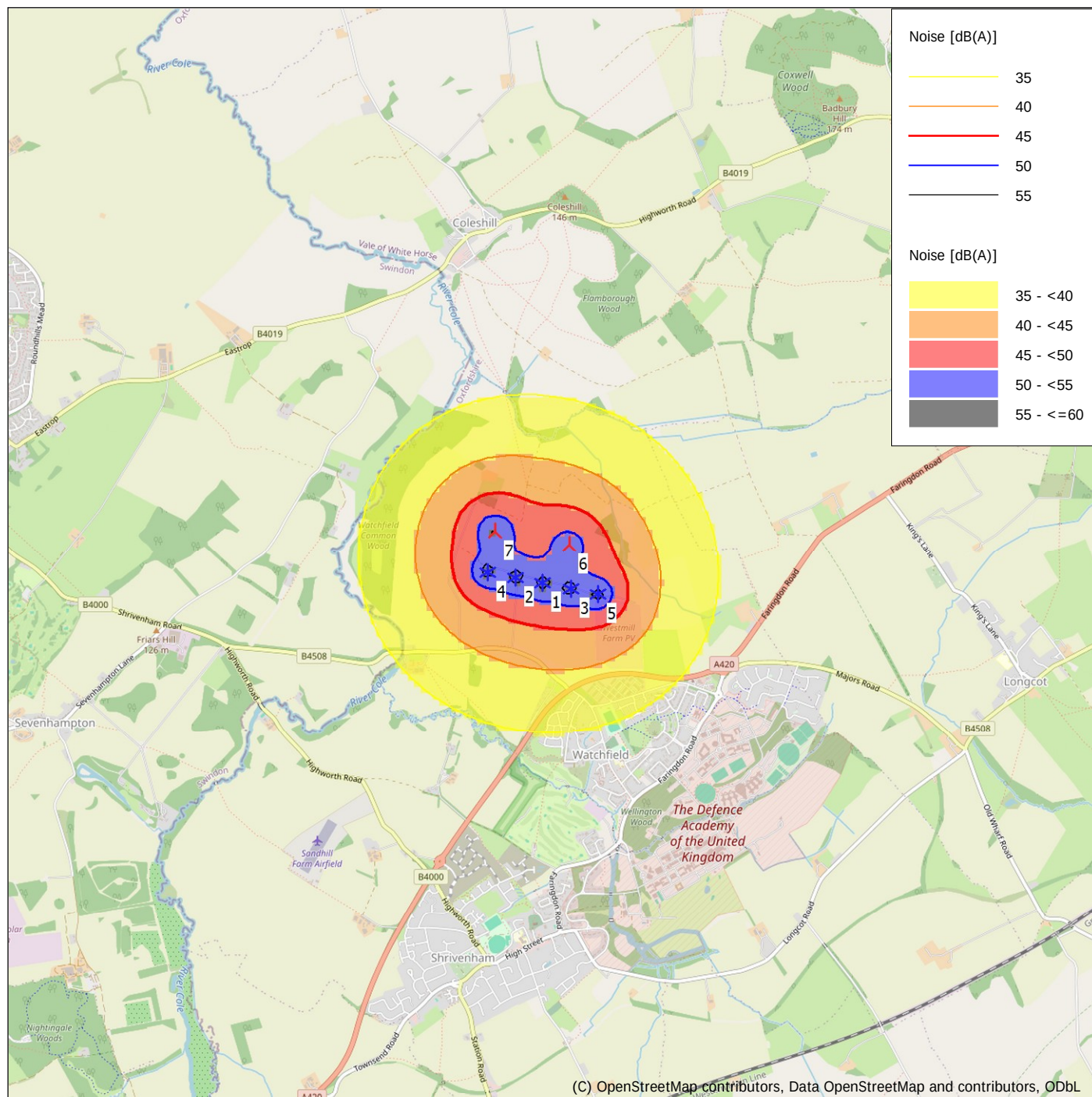
New WTG

Existing WTG

Noise calculation model: ISO 9613-2:2024 United Kingdom. Wind speed: 7.0 m/s
Height above sea level from active line object

DECIBEL - Map 8.0 m/s

Calculation: 5 Old WTG and 2 New WTG



0 500 1000 1500 2000 m

Map: EMD OpenStreetMap, Print scale 1:40,000, Map center British TM-OSGB36/Airy (GB/IE) East: 424,113 North: 191,527

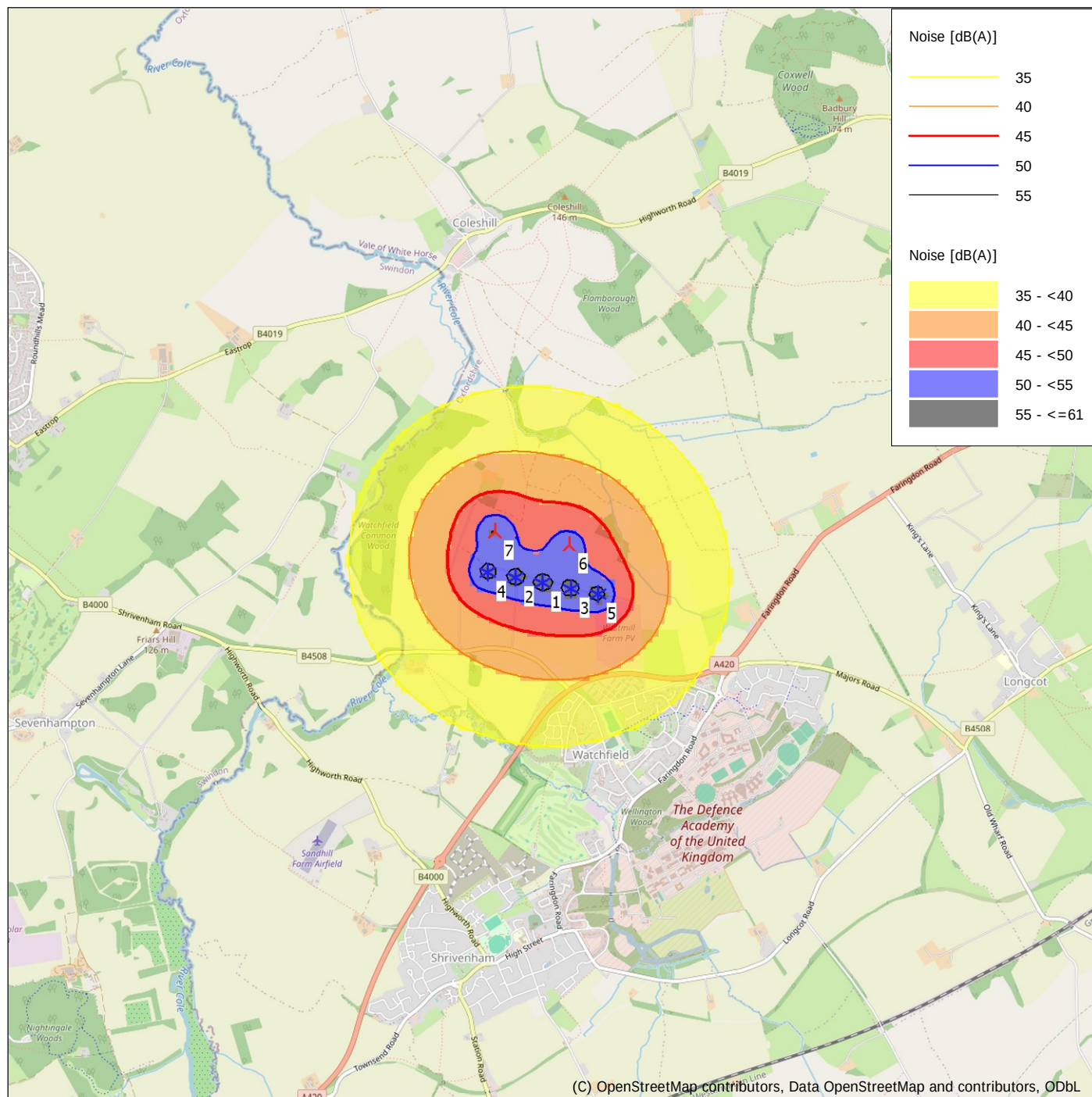
New WTG

Existing WTG

Noise calculation model: ISO 9613-2:2024 United Kingdom. Wind speed: 8.0 m/s
Height above sea level from active line object

DECIBEL - Map 9.0 m/s

Calculation: 5 Old WTG and 2 New WTG



(C) OpenStreetMap contributors, Data OpenStreetMap and contributors, ODbL

0 500 1000 1500 2000 m

Map: EMD OpenStreetMap, Print scale 1:40,000, Map center British TM-OSGB36/Airy (GB/IE) East: 424,113 North: 191,527

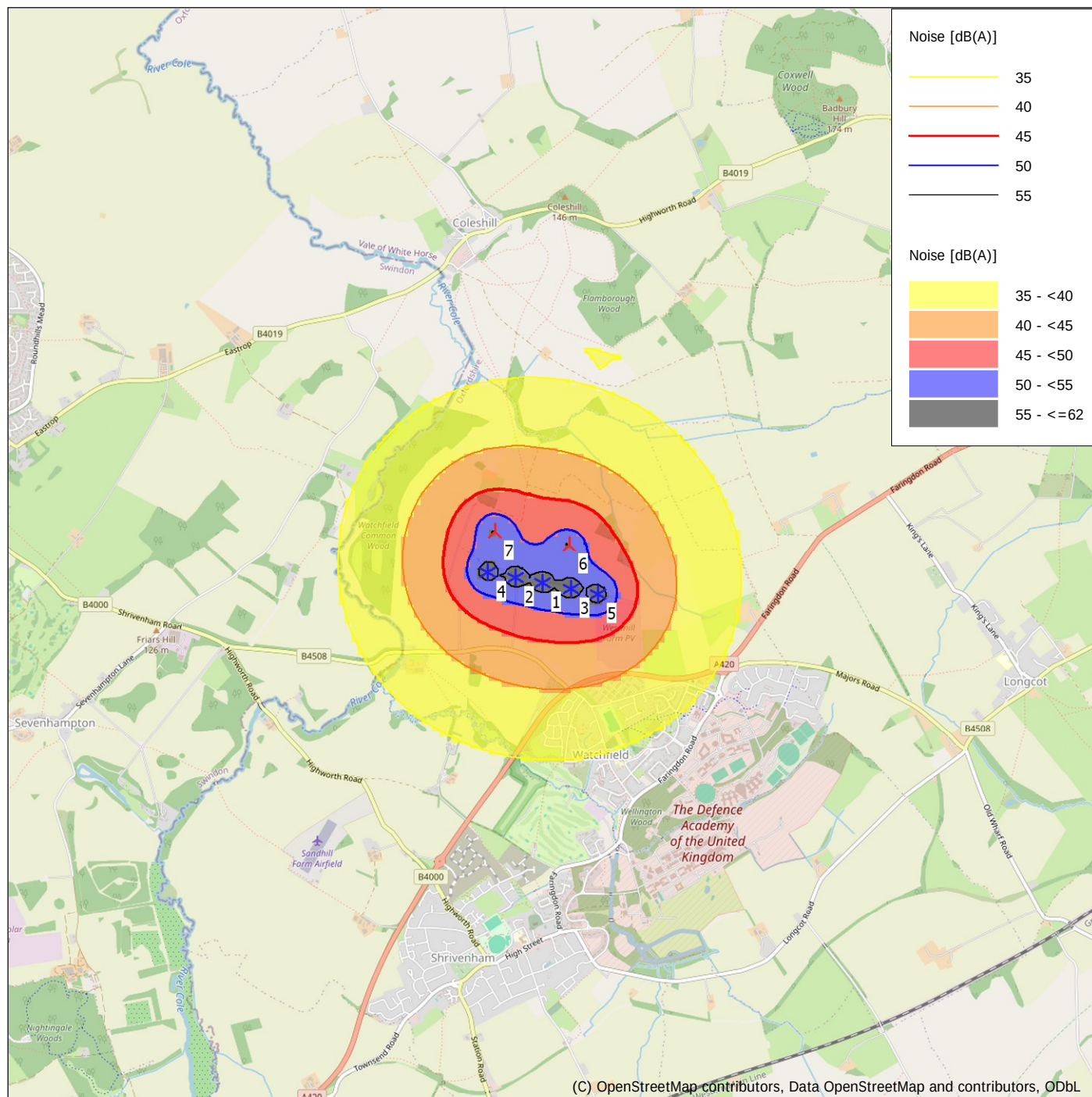
New WTG

Existing WTG

Noise calculation model: ISO 9613-2:2024 United Kingdom. Wind speed: 9.0 m/s
Height above sea level from active line object

DECIBEL - Map 10.0 m/s

Calculation: 5 Old WTG and 2 New WTG



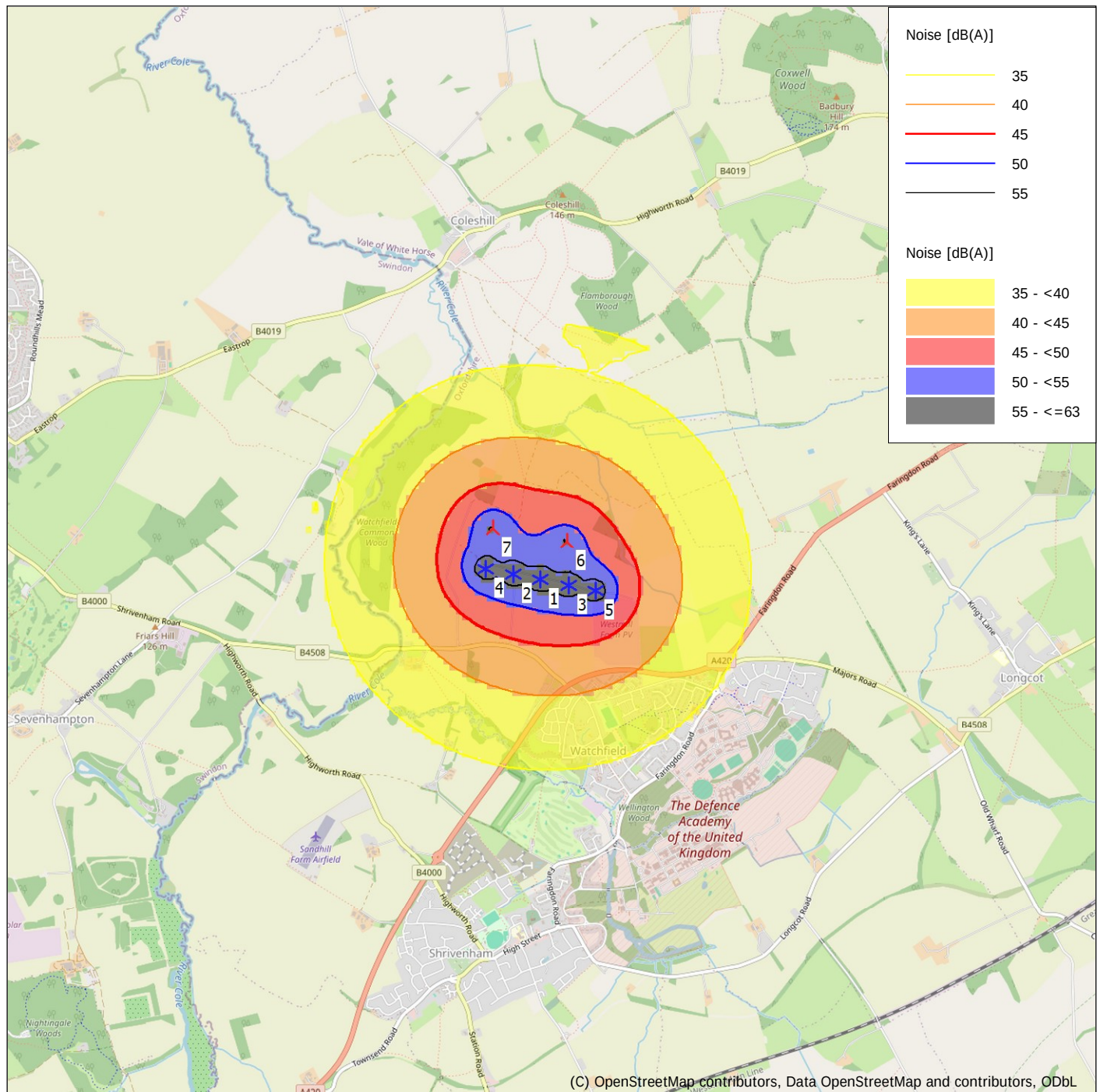
New WTG

Map: EMD OpenStreetMap, Print scale 1:40,000, Map center British TM-OSGB36/Airy (GB/IE) East: 424,113 North: 191,527

Noise calculation model: ISO 9613-2:2024 United Kingdom. Wind speed: 10.0 m/s
Height above sea level from active line object

DECIBEL - Map 11.0 m/s

Calculation: 5 Old WTG and 2 New WTG



0 500 1000 1500 2000 m

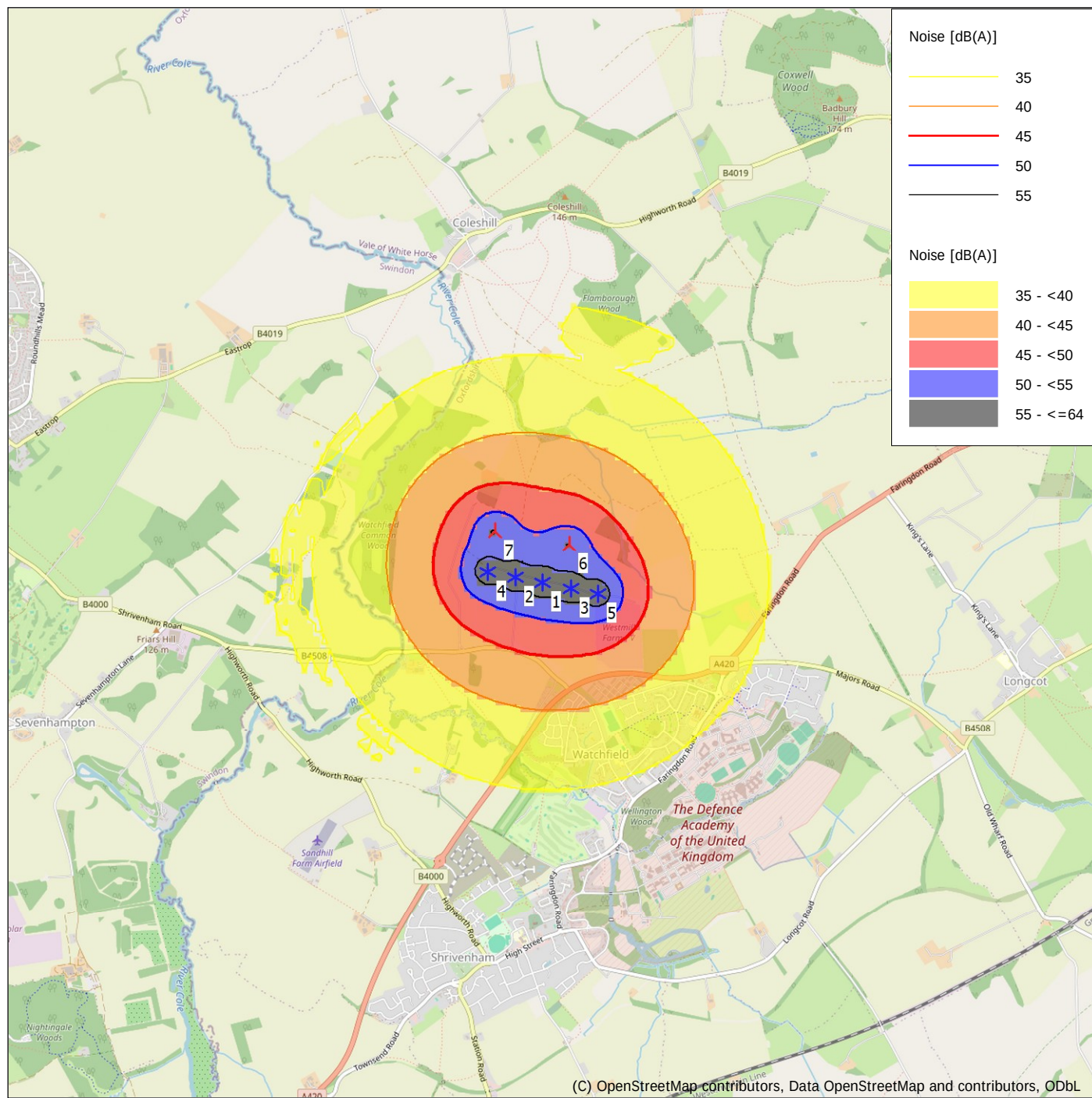
Map: EMD OpenStreetMap, Print scale 1:40,000, Map center British TM-OSGB36/Airy (GB/IE) East: 424,113 North: 191,527

New WTG Existing WTG

Noise calculation model: ISO 9613-2:2024 United Kingdom. Wind speed: 11.0 m/s
Height above sea level from active line object

DECIBEL - Map 12.0 m/s

Calculation: 5 Old WTG and 2 New WTG



0 500 1000 1500 2000 m

Map: EMD OpenStreetMap, Print scale 1:40,000, Map center British TM-OSGB36/Airy (GB/IE) East: 424,113 North: 191,527

⚓ New WTG

⚙ Existing WTG

Noise calculation model: ISO 9613-2:2024 United Kingdom. Wind speed: 12.0 m/s
Height above sea level from active line object